

Credit Risk PD Model Validation and Stress Testing

Educational project simulating a model validation analyst workflow for a credit risk probability of default model.

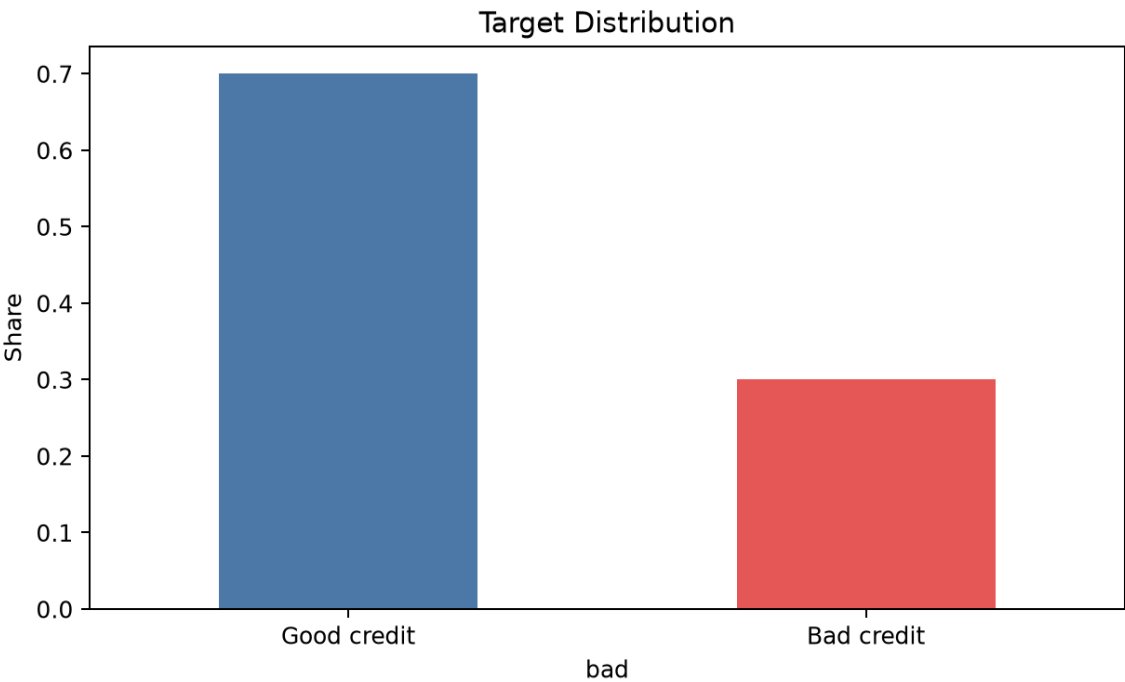
Item	Summary
Dataset	UCI Statlog German Credit Data
Sample size	1000 borrowers
Bad rate	30.0%
Best model selected	Logistic Regression
Regulatory framing	SR 11-7-style validation; Basel-style PD/LGD/EAD expected loss mapping.

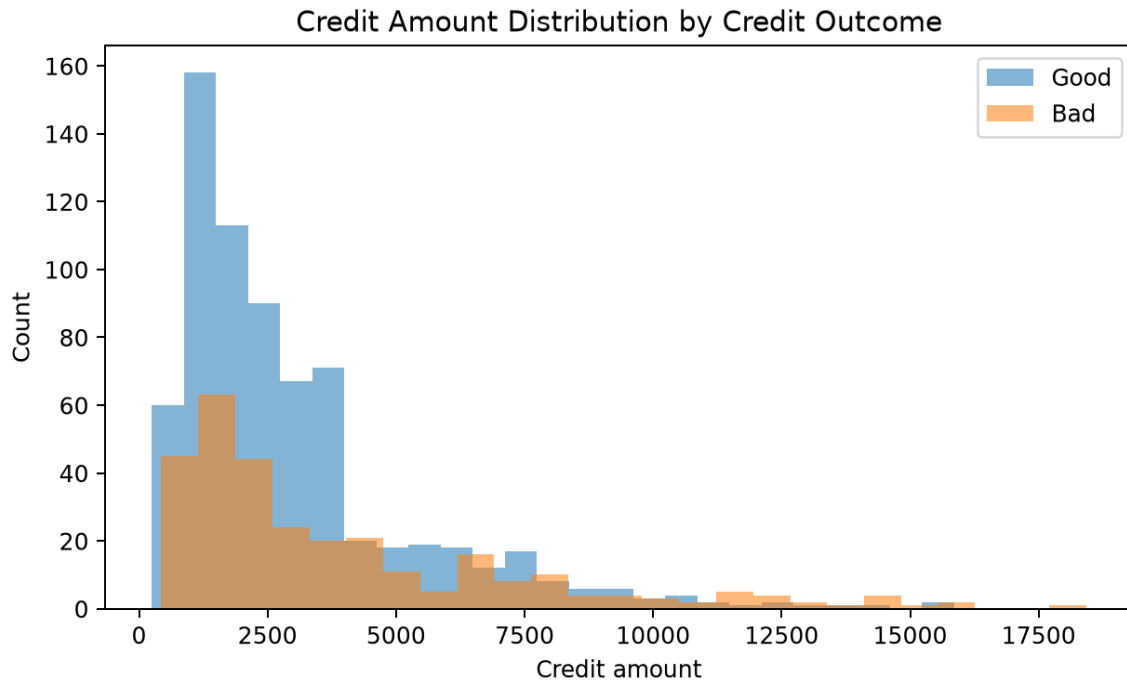
1. Model Purpose and Intended Use

The model estimates borrower-level probability of default (PD) for a small retail credit portfolio. It is not a production underwriting model; it is an educational model validation project designed to demonstrate quantitative testing, benchmarking, stress testing, and documentation discipline.

2. Data Quality Review

The UCI German Credit dataset contains 1,000 borrower records with categorical and numeric credit attributes. The validation review covers target distribution, variable reasonableness, borrower profile distributions, and observed risk patterns by loan duration and credit amount.





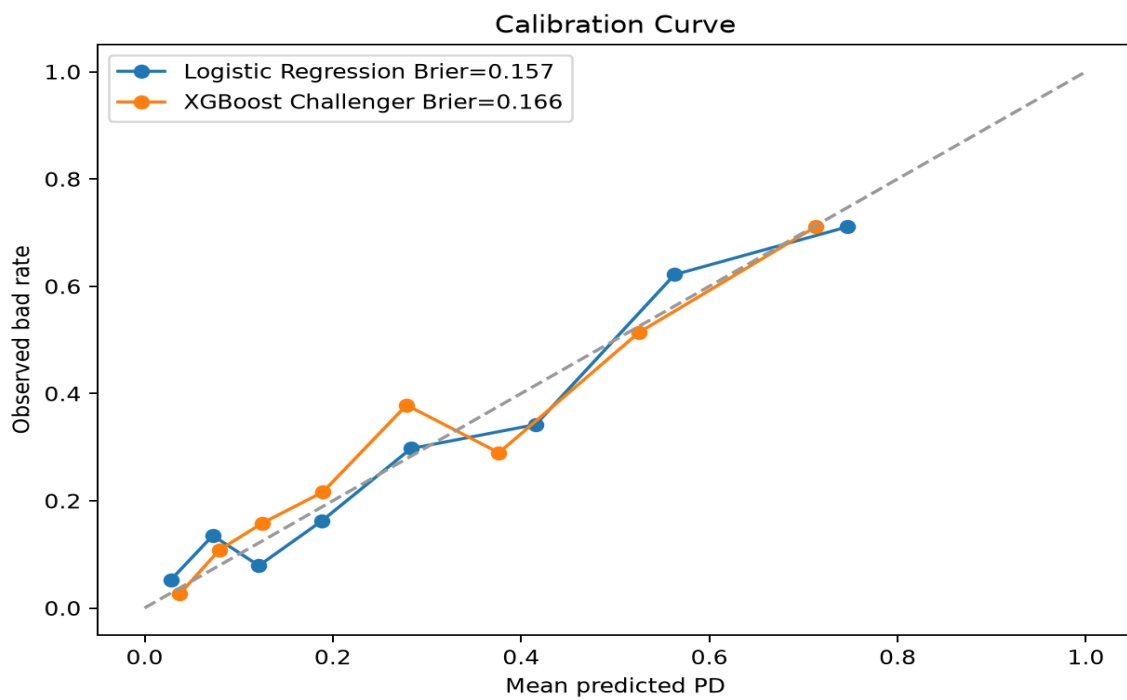
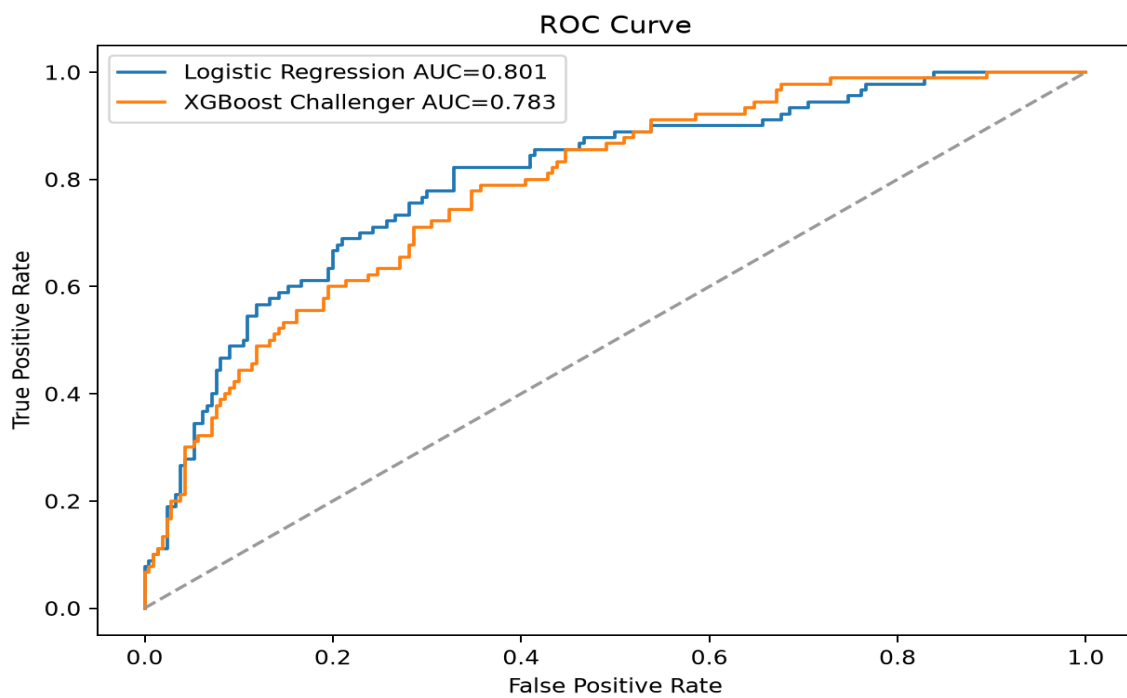
3. Methodology Review

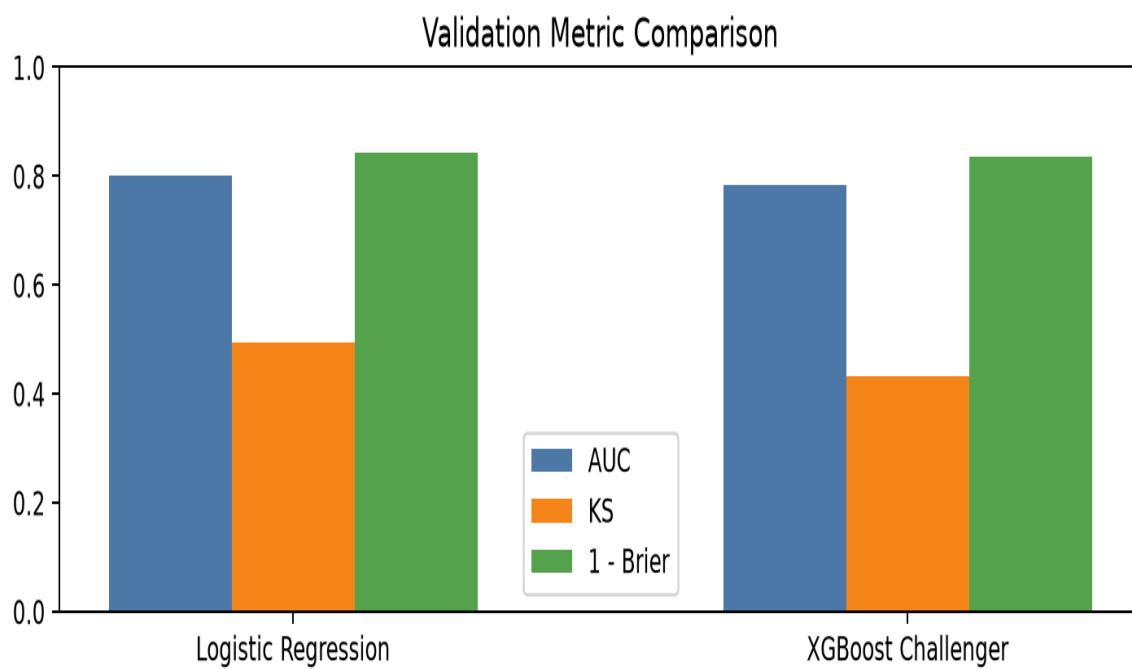
Two models are developed for benchmarking. Logistic Regression is used as an interpretable baseline aligned with traditional credit scoring practice. XGBoost is used as a nonlinear challenger model to test whether tree-based interactions improve discrimination. Both models use identical train/test split logic and preprocessing.

4. Outcomes Analysis and Quantitative Testing

Performance is evaluated through discrimination, calibration, cross-validation stability, and cost-sensitive threshold selection. AUC and KS measure ranking power; Brier Score and calibration curves measure probability quality; the German Credit cost matrix is used to reflect the higher cost of approving bad borrowers.

Model	AUC	KS	Brier	CV AUC mean	CV AUC std	Cost threshold	Expected cost
Logistic Regression	0.8011	0.4937	0.1574	0.7640	0.0241	0.2450	0.4967
XGBoost Challenger	0.7834	0.4317	0.1659	0.7713	0.0354	0.1400	0.5100

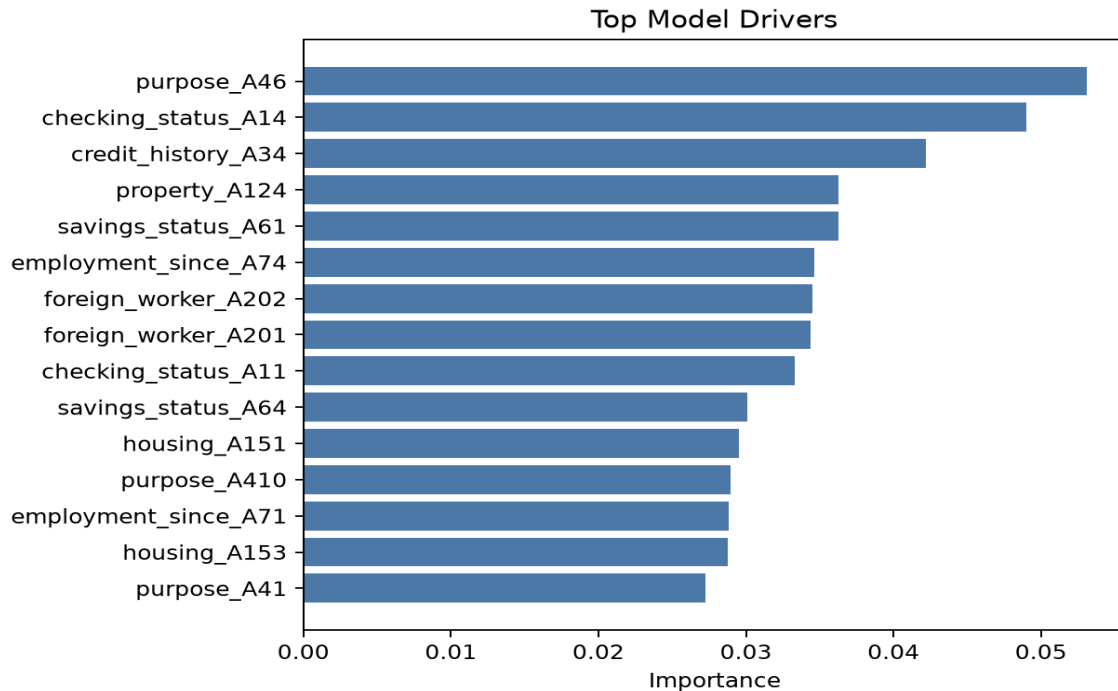




5. Model Drivers and Explainability

The most influential model drivers are reviewed for plausibility. In a real validation setting, this review would be supplemented with policy review, SME challenge, adverse action reason-code testing, and fairness controls.

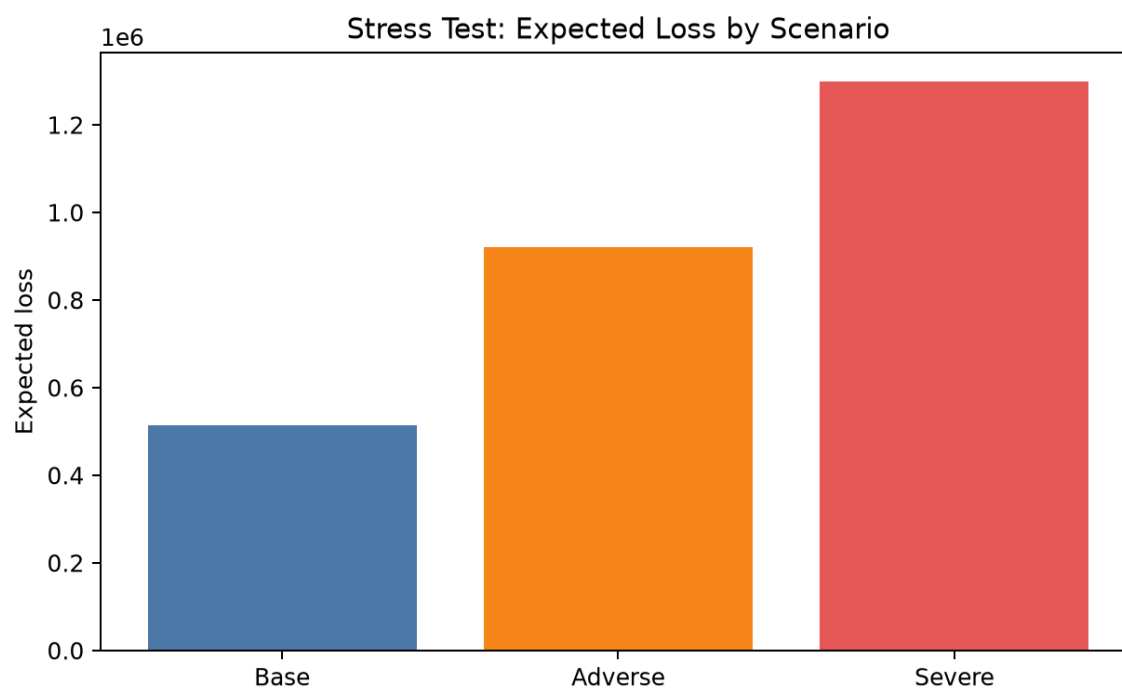
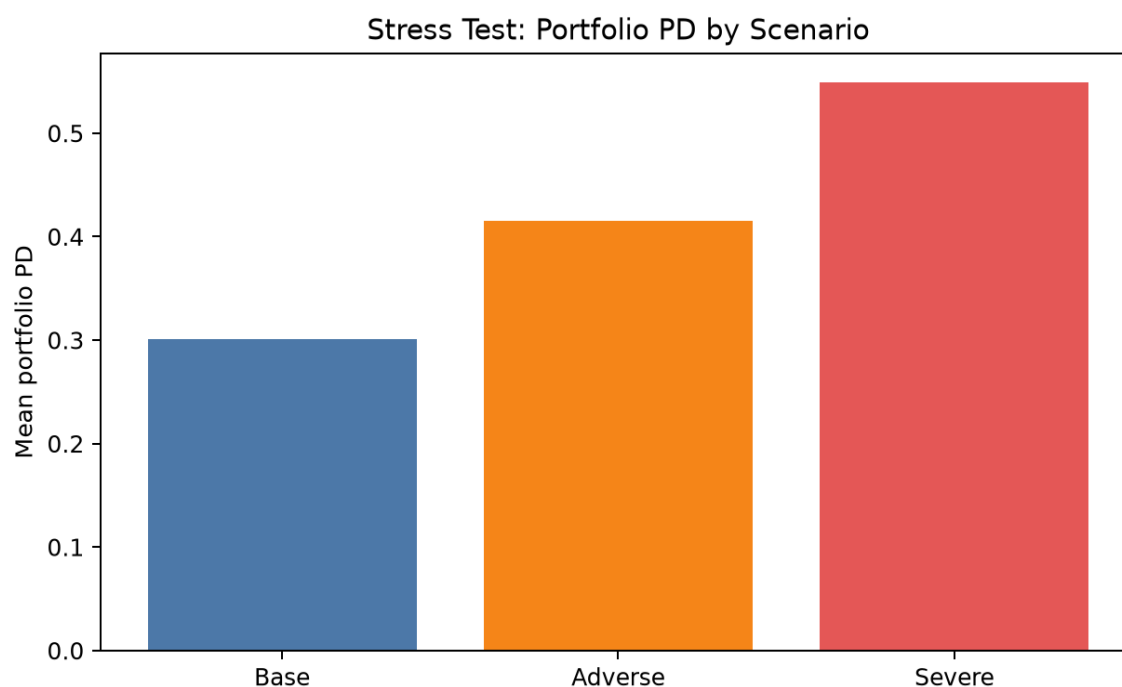
Feature	Importance
purpose_A46	0.0531
checking_status_A14	0.0490
credit_history_A34	0.0422
property_A124	0.0363
savings_status_A61	0.0362
employment_since_A74	0.0347
foreign_worker_A202	0.0345
foreign_worker_A201	0.0344
checking_status_A11	0.0333
savings_status_A64	0.0301



6. Stress Testing and Basel-style Expected Loss Mapping

Stress scenarios are designed to mimic adverse borrower and portfolio conditions. The project maps PD to expected loss using $EL = PD \times LGD \times EAD$, with LGD fixed at 45% and EAD proxied by credit amount. This is a simplified Basel-style risk parameter exercise, not a regulatory capital model.

Scenario	Mean PD	Portfolio EAD	Expected Loss	PD Uplift vs Base	EL Uplift vs Base
Base	0.3008	3,271,258	514,878	0.0000	0.0000
Adverse	0.4156	3,440,094	921,262	0.3818	0.7893
Severe	0.5491	3,896,741	1,298,799	0.8257	1.5225



7. Validation Findings and Limitations

- Finding 1: The challenger model should not be selected purely on AUC; calibration, cost-sensitive decisioning, interpretability, and stability should be considered together.
- Finding 2: Sample size is limited and may not represent a current banking portfolio. External validation and out-of-time testing are required before real use.
- Finding 3: Stress scenarios are educational and judgment-based. A production stress testing program would require macroeconomic scenario design, governance, and back-testing.

- Finding 4: Expected loss uses fixed LGD and credit amount as EAD. Production Basel models would separately estimate or validate PD, LGD, EAD, and maturity assumptions.

8. Recommendations

- Use Logistic Regression as the challenger/reference model for interpretability and governance comparison.
- Introduce out-of-time validation and population stability monitoring if new vintages become available.
- Calibrate PD estimates before using them for expected loss or decision thresholds.
- Document intended use, data limitations, assumptions, and model owner responses in a validation tracker.